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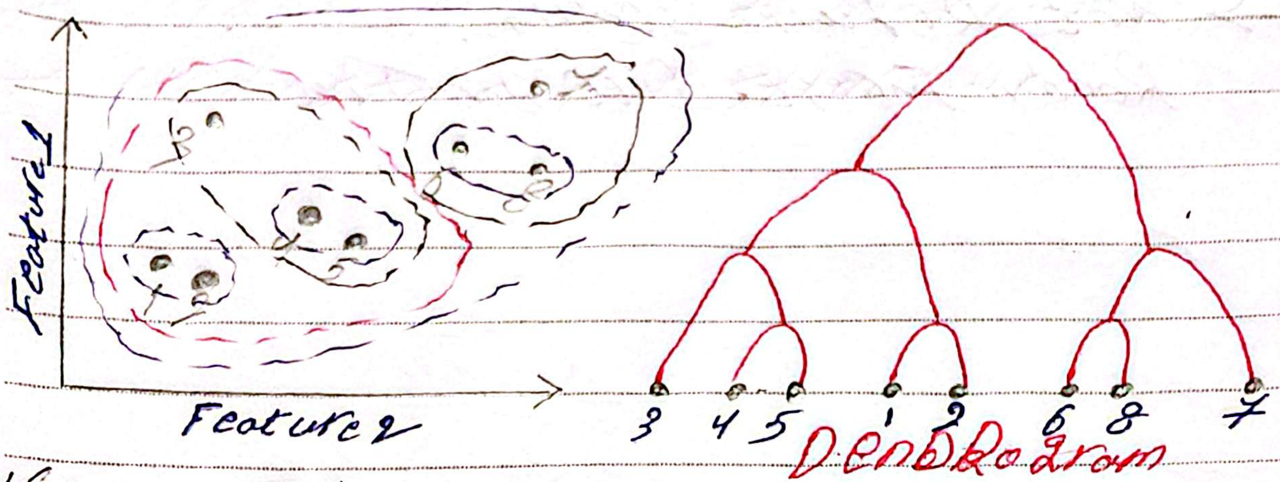
Hierarchical Clustering

→ It is an supervised machine learning that groups data into a tree-like structure (hierarchy)



* Core Idea:

→ Build a hierarchy (tree) of clusters

→ Cut the tree at certain level to get the desired number of clusters

* Types of Hierarchical Clustering:

[1] Agglomerative (Bottom-up) _ most common

→ Start with each data point as its own cluster

→ Merge the closest clusters step by step

→ do it until all points are in one cluster

[2] Divisive (Top-Down)

→ Start with all data in one cluster

→ Split it recursively into small clusters

Linkage Methods:

* Single Linkage:
  Distance = Closest Pair of Points
  → Can Create Chain-like Clusters

* Complete Linkage:
  Distance = farthest Pair of Points
  → Produces Compact Clusters

* Average Linkage:
  Distance = average between all Pairs
  → Balanced approach

* Ward's Method (Most Important)
  Minimizes Variance within Clusters
  → Often gives the best results.

ADVANTAGES:
→ Resulting hierarchical representation can be very informative
→ Provides an additional ability to visualize

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DisADVantages:

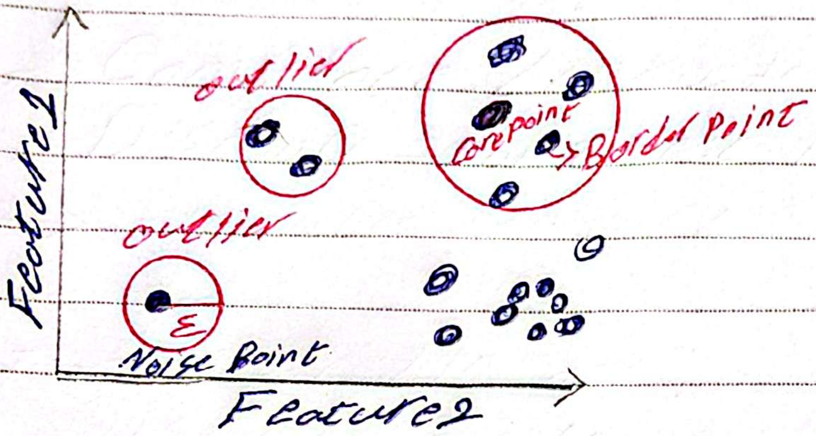
Sensitive to Noise and outliers.

→ when it use:

* You don't know the number of clusters
* Dataset is small to medium.
* understand datastructure.

DBSCAN - Density-Base Clustering

→ it is unsupervised learning algorithm that groups data based on density (How Packed Points are) rather than Just distance.



ε: Radius around a Point
MinPts: Minimum number of Points inside ε

Advantage: * Automatically detects outliers
* Can find arbitrary-shaped Clusters (Not just circles)

	K-mean	Hierarchical	DBscan
Need K	✓	✗	✗
Detect outliers	✗	✗	✓
cluster shape	spherical	flexible	Any shape
speed	Fast	slow	medium

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